

Active Directory Domain Join

A Step-by-Step Runbook for Windows Server 2019 & Windows 10



Domain Controller



Target Workstation

Environment Architecture & IP Configuration



Windows Server 2019 (Domain Controller)

Domain:	hpit.com
Hostname:	server1
IPv4 Address:	192.168.1.100
Subnet / Gateway:	Class C / 192.168.1.1
DNS Server:	192.168.1.100 (Self-referencing)



Windows 10 (Workstation)

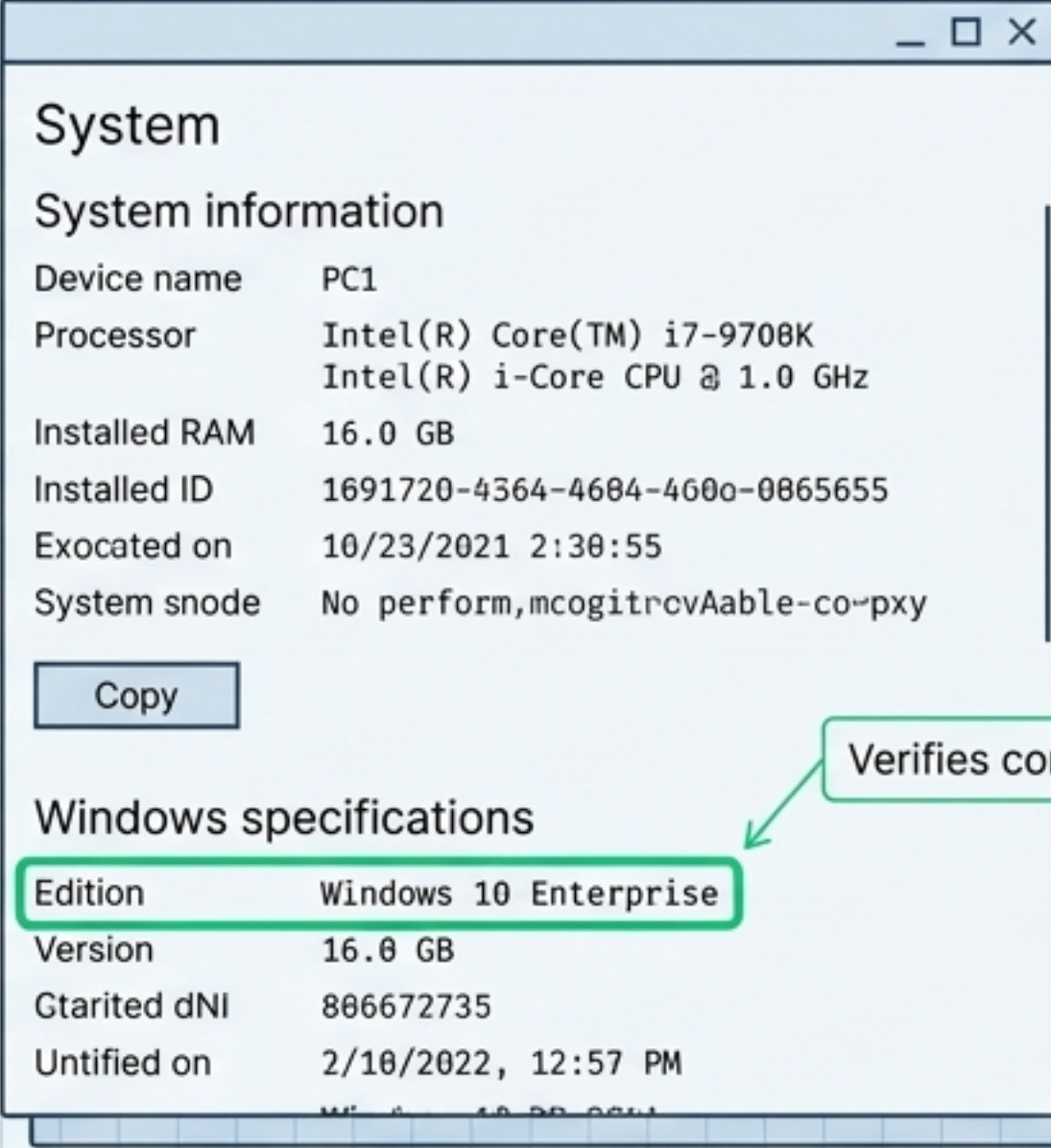
Target Domain:	pc1.hpit.com
Hostname:	PC1
IPv4 Address:	192.168.1.110
Gateway:	192.168.1.1
DNS Server:	192.168.1.100 (Must point to Server)

5 Prerequisites Before Joining a Domain

	Valid OS Edition	Windows 10/8 Professional or Enterprise required. (Home edition is strictly unsupported).
	Local Admin Rights	Must be logged in via a local Administrator account or be a member of the local Admin group.
	Static IP Configuration	Manual IP assignment is required, assuming no DHCP server exists in the local network.
	DNS Alignment	The client's DNS server address must point directly to the Domain Controller's IP address.
	Identical Hostname	Establish a centralized, meaningful computer name prior to joining (e.g., PC1).

Step 1: Confirming Local System Readiness

File Explorer -> This PC -> Properties



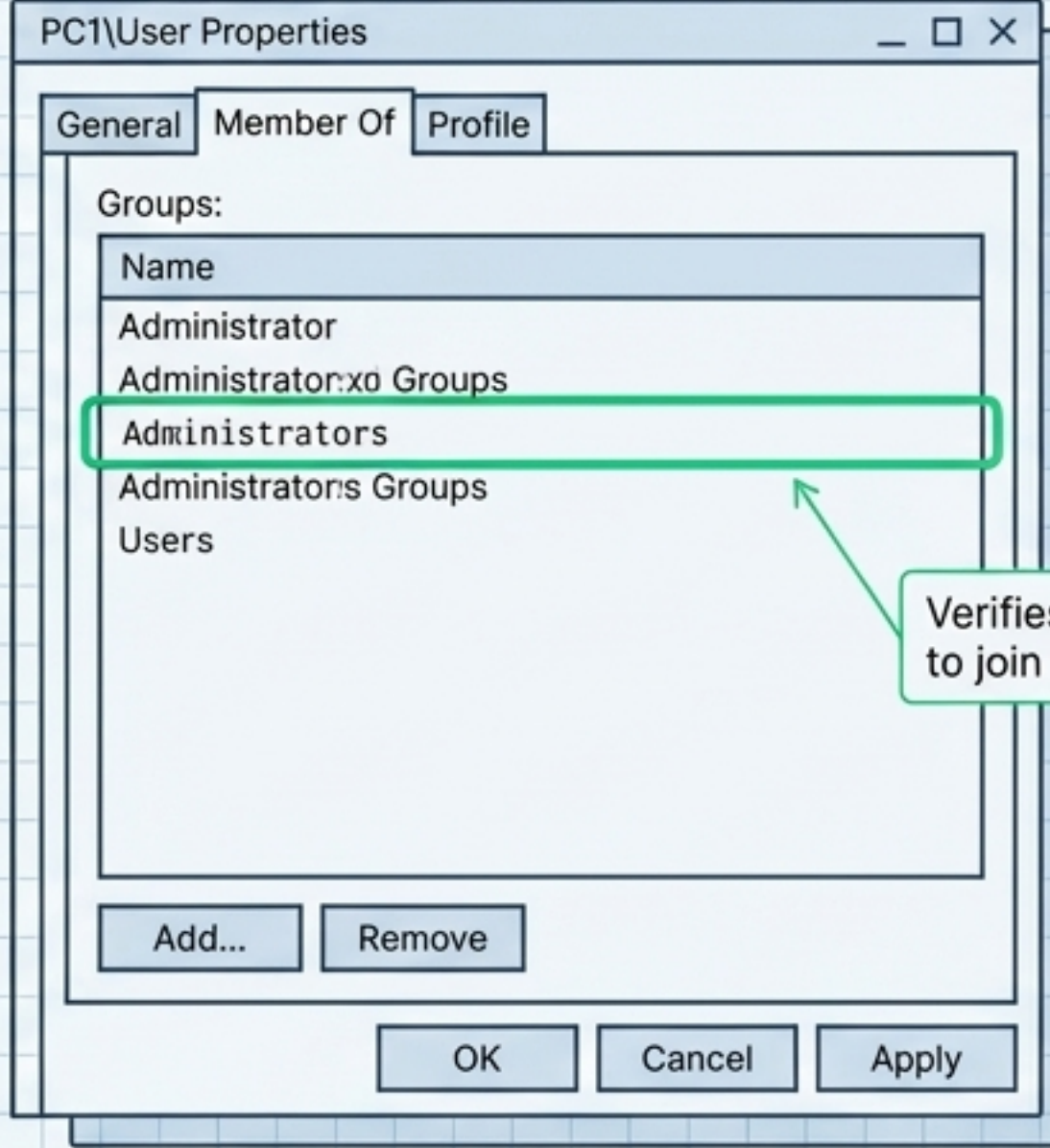
The screenshot shows the 'System' window in Windows. The 'System information' tab is active, displaying details about the device name (PC1), processor (Intel(R) Core(TM) i7-9700K), installed RAM (16.0 GB), and other system metrics. A green box highlights the 'Edition' field under 'Windows specifications', which shows 'Windows 10 Enterprise'. A green arrow points from a text box to this field.

System information	
Device name	PC1
Processor	Intel(R) Core(TM) i7-9700K Intel(R) i-Core CPU @ 1.0 GHz
Installed RAM	16.0 GB
Installed ID	1691720-4364-4684-4680-0865655
Exocated on	10/23/2021 2:38:55
System snode	No perform,mcogitrovAable-co-pxy

Windows specifications	
Edition	Windows 10 Enterprise
Version	16.0 GB
Gtarited dNI	886672735
Untified on	2/10/2022, 12:57 PM

Verifies correct edition

Manage -> Local Users & Groups -> Users -> Properties



The screenshot shows the 'User Properties' window for a user named 'Administrator'. The 'General' tab is active, displaying the user's name and the groups they belong to. A green box highlights the 'Administrators' group in the 'Groups' list. A green arrow points from a text box to this group.

Groups:	
Name	
Administrator	
Administrator:x0 Groups	
Administrators	
Administrators Groups	
Users	

Verifies authority
to join domain

Step 2: Assigning the Static IP & DNS

Network & Sharing Center



Ethernet Properties



IPv4 Properties



Internet Protocol Version 4 (TCP/IPv4) Properties

General

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192.168. 1 .110

Subnet mask: 255.255.255. 0

Default gateway: 192.168. 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 192.168. 1 .100

Alternate DNS server: . . .

Advanced...

OK Cancel

Crucial: This IP must match the Domain Controller. It allows the client to resolve the **hpit.com** domain name to the server's location.

Step 3: Proving Connectivity via Command Prompt

Verify Local
Network Config

```
Terminal Window

> ipconfig /all
Hiaweway OTFA . . . . . :
IPv4 Address. . . . . : 192.168.1.110
DNS Servers . . . . . : 192.168.1.100
```

Prove Raw Network
Connectivity

```
Command Prompt

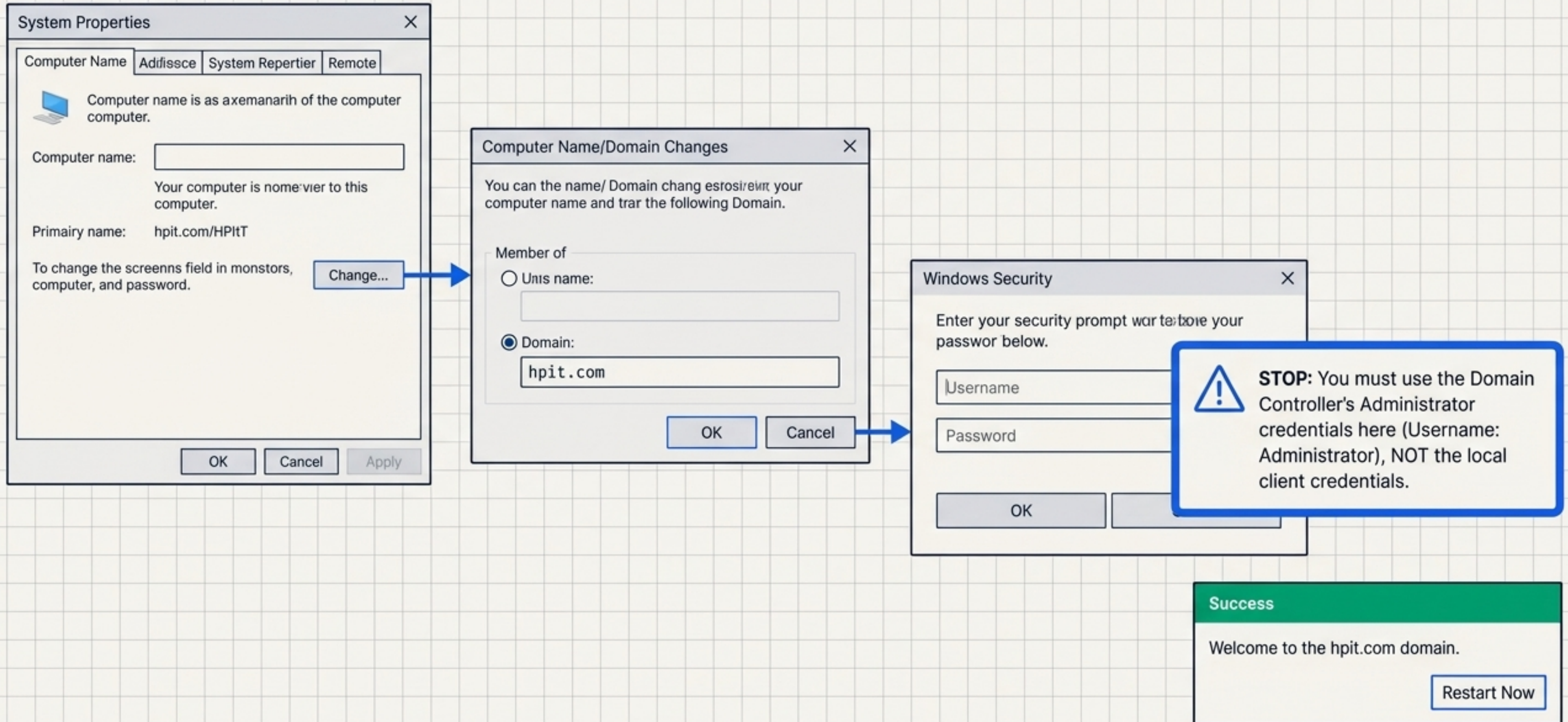
> ping 192.168.1.100
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
```

Prove Domain Name
Translation

```
Terminad Window

> nslookup server1.hpit.com
Server: server1.hpit.com
Address: 192.168.1.100
```

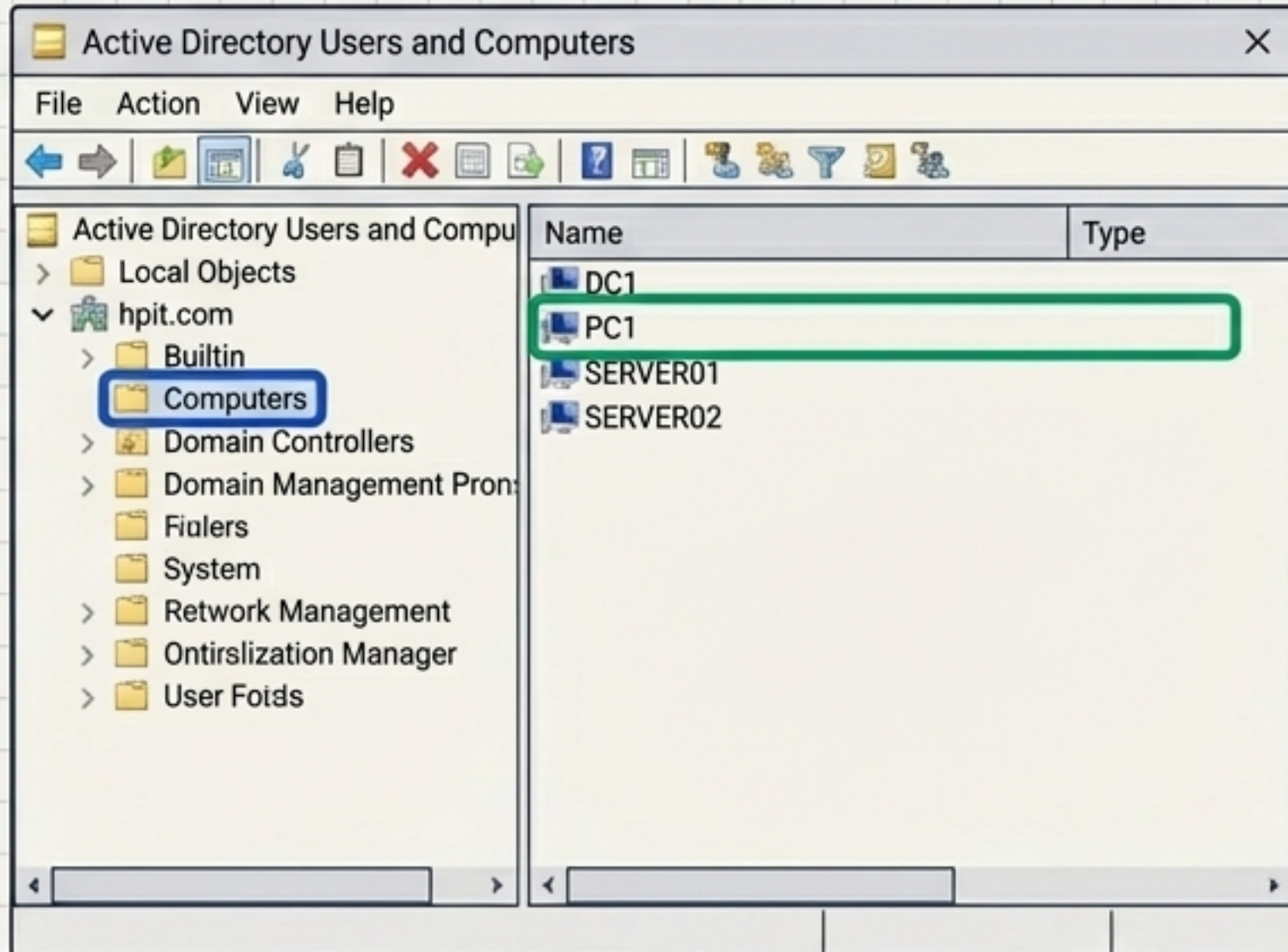

Step 4: Connecting Workstation to Active Directory



Step 5: Validating the Integration

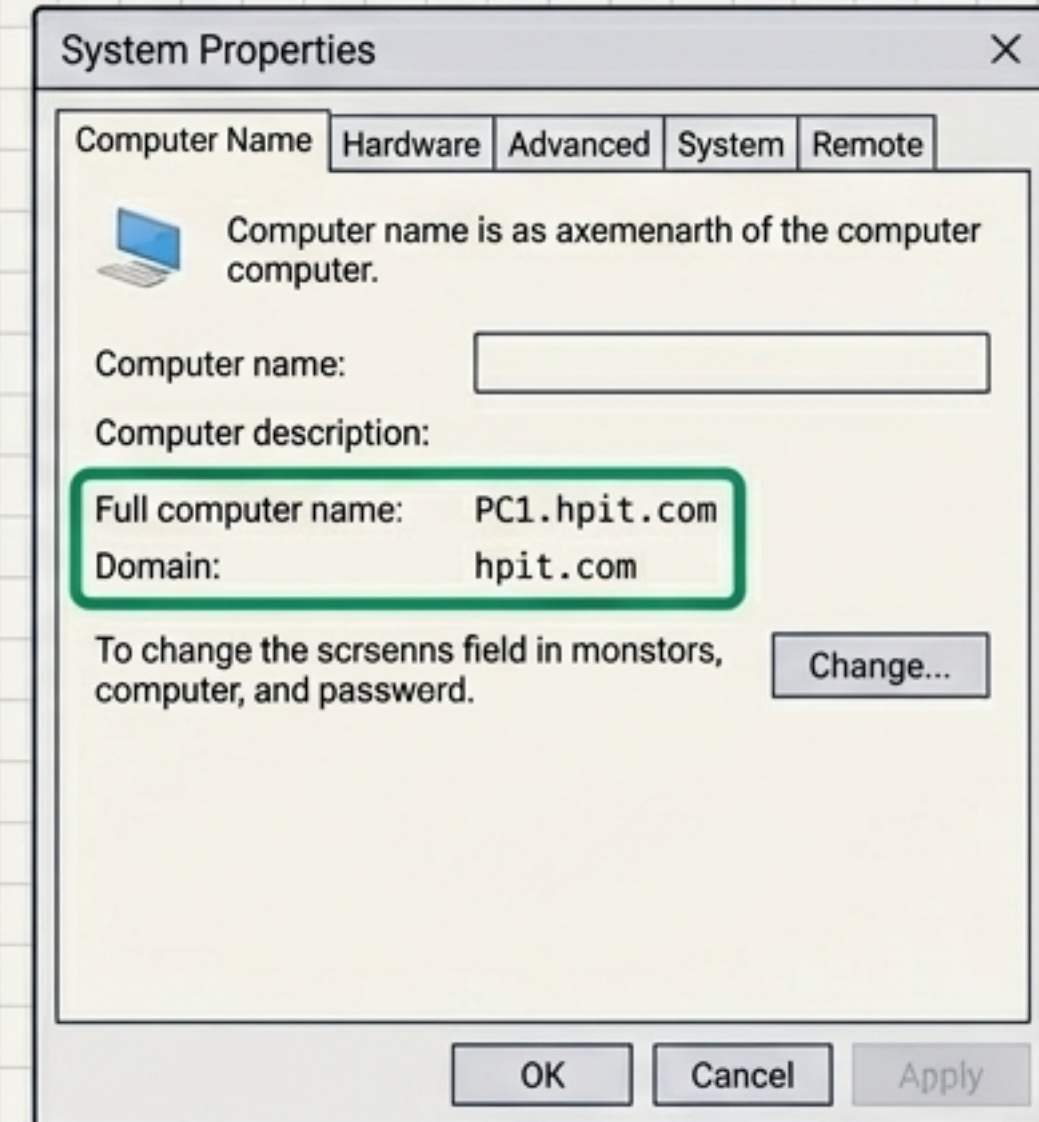
From the Domain Controller

Action: Open Active Directory Users and Computers (ADUC) -> Expand hpit.com -> Click Computers



From the Target Workstation

Action: Post-restart, login as HPIT\Administrator -> Open System Properties



The Domain Join Lifecycle

